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Design of an Optimal ANFIS Traffic Signal Controller by using Cuckoo Search for an Isolated Intersection

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***Abstract*—An optimal design of Adaptive Neuro-Fuzzy Inference System (ANFIS) traffic signal controller is presented in this paper. The proposed controller aims to adjust a set of green times for traffic lights in a single intersection with the purpose of minimizing travel delay time and traffic congestion. The ANFIS controller is trained, to learned how to set green times for each traffic phase. This intelligent controller uses the Cuckoo Search (CS) algorithm to tune its parameters during the learning pried. Evaluating the performance of the proposed controller in comparison with the performance of a FLS controller (FLC) with predefined rules and membership functions, and also three fixed-time controllers, illustrates the better performance of the optimal ANFIS controller against the other benchmark controllers.**

I. Introduction

Increased travel demands in urban areas needs appropriate management. Traffic signal control is one of the solutions that has influence on traffic congestion.

Researches have been focused on a range of methods to control traffic signal timing, including the three generations of traffic signal control methods. The first generation fixed-time method requires pre-set signal sequences. The second generation sets the signal timing based on received information from traffic detectors. The third generation is characterized by the ability of dynamic decision making and distributed control.

The traditional traffic controlling methods cannot handle today's traffic condition of fluctuating and ever increasing volumes. Traffic flows in urban area are vague, complex, random and fuzzy. In this situation it is difficult to propose a pre-defined formula for traffic controlling [1]. Artificial Intelligence (AI) techniques, which have the ability of learning and decision making, are effective to control traffic congestion at an intersection [2].

AI methods are self-organizing and replay to dynamic changes of conditions. Learning from experience is one of the characteristics of AI methods that gives them the ability to address real world problems. Studies have been done to apply these methods to improve the performance of the controlling [9]–[12]. FLS is one of these methods that is suitable for analysis and modeling of process in which human approach is characterized by rough approximation [15].

Using fuzzy theory instead of crisp set theory gives the ability to implement the real-world scenarios in more details. FLS is suitable for managing problems with linguistic term information. Through this method expert's knowledge can be included during the design. An FLS maps the inputs to the output of the system. The domain of a problem with just two zero and one value is extended to the range of whole numbers between zero and one by fuzzy set. Membership functions are used to show the degree of dependency to each fuzzy set. An input or output value may belongs to more than one fuzzy set.

For the first time FLS is applied for controlling traffic signals timing in 1977 [15]. The designed controller has three inputs and one output and it was for a two-phase single intersection. The traffic network model used random vehicle arrivals and no turning movements. In this research, the fuzzy rules were developed to evaluate and make decision about the appropriate extension or termination of current green phase. It was assumed that the vehicle detectors were placed upstream from the intersection to inform the controller about the arrival of future vehicles. Evaluating the system was done by comparing with the efficient vehicle-actuated method and the result of simulation showed the better performance of designed FLC.

Researchers in [16] applied FLS to control a single two-way streets intersection. In this work, the number of vehicles that had already passed the intersection and the length of the queue at the stop lights are received as inputs and the amount of extension of green time was the output of the FLC. Additional strategies were considered for adapting the numerical bounds on the input and output.

Nair and Cai [26], also proposed a FLC for a single intersection. They aimed to ensure smooth flow of traffic and reduce the delay time. In design of FLC, it is usually attempted to optimize the performance of the network by maximizing traffic flows or minimizing traffic delays under typical traffic conditions, consequently, these controllers are not the optimal for exceptional traffic cases such as road accidents. Nair and Cai designed an FLC for both normal and exceptional traffic conditions. Their proposed system had different traffic detector sensors placed at both incoming and outgoing links. The controller made optimal decisions by utilizing the information received from detectors. Results showed that the performance of their designed traffic controller was similar to that of conventional fuzzy traffic controllers under normal traffic conditions and was better

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than the others under abnormal traffic conditions. Many other researchers are also proposed FLC for a single intersection [33]– [35].

For the first time FLS was used in a multi-intersection network in the beginning of 1990. Chiu and Chand [36], presented a fully distributed system with cooperative local controllers to self-organizing traffic signal control. In order to fix the parameters of each controller, the local traffic conditions and the parameters of the adjacent intersections are considered . Each local controller used a set of fuzzy rules to set the signal timing parameters. Their system, adjusted cycle time, phase split and offset. The proposed approach provides the ability for the local controllers to have their own cycle time when the coordination between intersections is not important.

Choy *et al.* [22] proposed a hybrid approach in their work. They used AI concepts to implement a cooperative, hierarchical, multiagent system. The proposed method was suitable for real-time traffic signal control of a large-scale traffic network. The issue of controlling the whole traffic network was divided to various subproblems and each managed with a controller (agent) by fuzzy neural decision making capability. This method is applied for controlling traffic signal timing in a section of the Central Business District of Singapore. The results of experiments declare reducing total vehicle stopping time by 50% and the total mean delay by 40% compared to real-time adaptive traffic control system of the moment.

Researchers in [27] and [28] applied type-2 FLS for controlling traffic signal lights. They mentioned that the nature of the urban traffic led to fuzzy type-2 become more appropriate to be employed in design of signal controllers. In [28], it was claimed that using AI-based methods such as neural network (NN) in design of signal controller, with a large training data set and all uncertainties they may contain, is not appropriate. The researchers developed a distributed architecture based on type-2 FLS. In their proposed structure, all agents were homogeneous and had similar decision making capabilities. Each agent calculated the suitable green time based on average flow rate, queue length, and communicated data from the immediate neighbors. Experiments results illustrated about 40% improvement against fixed-time method.

Rich literature traffic signalling using FLS represent its high performance for traffic applications. FLC in most of these works makes decisions about extending or terminating of the green time in the current phase. We have presented a review on many of the related works to this field in [3].

In this paper, FLS is used to control traffic congestion by allocating suitable time to traffic signal phases. In this regard, two FLCs are implemented. One of them uses predefined rules and membership function in its fuzzy sets and called fixed-FLC. This controller is used as a benchmark. The other one is an adaptive neuro fuzzy controller that applies Cuckoo Search, a new nature inspired optimization method, to optimally adjust the controller's rule base. This controller named CS-ANFIS calculates the average delay of the intersection after each round of simulation and update the parameters to reach the optimal controller. In both of fuzzy based proposed controllers we can have flexible traffic cycle time. Before start of each phase the appropriate green times for that phase is estimated and sent to the controller. In addition, three fixed-time controllers are also designed as benchmarks.

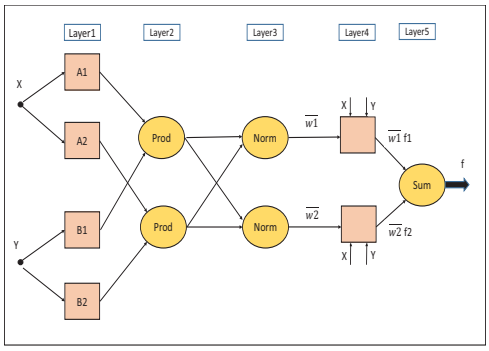


Fig. 1. ANFIS architecture

The rest of this paper is organized as follows. The related background is discussed in section II. In section III, our designed FLCs are introduced. Section IV presents experimental results and discussion and finally conclusion are listed in section V.

II. Related Background

A. Adaptive Neuro-Fuzzy Inference System

ANFIS is composed of the combination of FIS and NN. It works by using NN rules to tune the parameters of the FIS. In a ANFIS model a first order Sugeno fuzzy system adaptively modifies the rules of the system to reach the optimal ones. A typical architecture of an ANFIS is presented in Fig. 1. Two types of nodes are presented in this model, circle indicates a fixed node and square is for an adaptive node. It is assumed that the examined FIS has two inputs and one output. For a first-order Sugeno fuzzy model, a typical rule set with two fuzzy "if then" rules can be expressed as follows Eq. 1 and Eq. 2:

$$Rule1 : If\ x\ is\ A_1\ and\ y\ is\ B_1,\ then\ f_1(x) = p_1x + q_1y + r_1 \tag{1}$$

$$Rule2 : If\ x\ is\ A_2\ and\ y\ is\ B_2,\ then\ f_2(x) = p_2x + q_2y + r_2 \tag{2}$$

where x and y are the crisp inputs, and A_i and B_i (for $i = 1, 2$) are the linguistic labels associated with the node function.

Fig. 1 has five layers. Description about each layer is presented below:

Layer 1 : Every node i in this layer is an adaptive node with a node function such as Eq. 3. ($O_{l,i}$ is the output of the ith node of layer l).

$$O_{1,i} = \mu\ A_i(x), \quad i = 1, 2 \tag{3}$$

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where x is the input to node i, and A_i is the linguistic label (small, medium, etc.) related to the current node function. In better words $O_{1,i}$ is the membership grade for each fuzzy set members (A_1, A_2, B_1, B_2).

Layer 2 : The outputs of this layer represent the firing strength of each rule Eq. 4. A minimum value of two input weights is selected in this layer.

$$O_{2,i} = \mu\ A_i(x) \times \mu\ B_i(x), \quad i = 1, 2 \tag{4}$$

Layer 3 : The outputs of this layer called normalized firing strengths. The ith node calculates the ratio of the ith rulet's firing strength to the sum of all rulet's firing strengths Eq. 5.

$$O_{3,i} = \bar{W}_i = \frac{w_i}{w_1 + w_2}, \quad i = 1, 2 \tag{5}$$

Layer 4 : Linear functions of input signals are calculated in this layer. Each node i in this layer is an adaptive node Eq. 6.

$$O_{4,i} = \bar{W}_i f_i = \bar{W}_i(p_i x + q_i y + r_i), \quad i = 1, 2 \tag{6}$$

where $\bar{W}_i$ is the normalized firing strength of *layer 3*, p_i, q_i, r_i are the parameters of this layer and reffered as *consequent parameters*.

Layer 5 : This layer is made of a single node labeled sum. The output of this layer is the overall output and summing all incoming signals Eq. 7.

$$overalloutput = O_{5,i} = \sum_i \bar{W}_i f_i = \frac{\sum_i \bar{W}_i f_i}{\sum_i \bar{W}_i} \tag{7}$$

B. Cuckoo search optimization algorithm via Lévy flight

Cuckoo search (CS) is a new heuristic search algorithm, developed by Yang and Deb in 2009 . In this method three different rules are considered:
First: Each cuckoo lays an egg, chooses a host nest and dispose it there.
Second: The best nest, nest with higher quality of eggs, will carry to the next generation.
Third: The number of host nests are fixed and the probability of egg-discovery by host bird is $p_a \in [0, 1]$. The host bird in CS can throw out the parasitic egg or abandon that nest for a better new nest. In fact, p_a is the probability of substituting the nest by a new nest which is a new random solution.

Lévy walk generates a substantial of the new solution around the best solution obtained so far, this is useful to speed up the local search. In addition, some of the new solutions should be generated far enough from the current solution. This group will be generated by far field randomization to assurance of not being trapped in a local optimum.

Yang and Deb' claim [4], there are some similarities between CS and hill-climbing compared to some large scale randomization. While, there are also significant differences: Firstly, CS is a population based algorithm similar to Genetic Algorithm (GA) and PSO, but it uses some sort of elitism and/or selection similar to harmony search. Secondly, CS is more efficient in randomization. In CS the step length is heavy tailed and any large step is possible. Thirdly, the number of parameters need to be tuned in CS are less than GA and PSO. The last item makes CS more appropriate for wider class of optimization problems.

III. Proposed Optimal CS based ANFIS Controller

One of the difficulties in applying FLS for a system is how to consider suitable rules to obtain the best efficiency of the FLC. In ANFIS system, the first order Sugeno model modifies the rules and adaptively learns to reach the optimal parameters for the controller.

In one of our previous work [5], a comparison between the performance of a fixed-time controller, fixed-fuzzy, and GA-FLC is presented. In GA-FLC the controller adapts its output membership function by using GA. This work, revealed the better performance of the optimal GA-FLC compared to the other controllers designed in the study.

In our proceeding work [6], we evaluated the performance of the proposed GA-ANFIS controller against GA-FLC. The results showed the premier performance of the GA-ANFIS. Regarding to the superiority of the CS in different field, we apply CS instead of GA in this work.

In this work, an ANFIS controller is designed for a four-way intersection. This controller has four inputs and one output. The number of vehicles on each approaching link of the four-way intersection makes the inputs of the controller. The output is the proposed green time for the next phase of the traffic cycle. After the end of each phase the traffic data is transfered to the ANFIS controller and the controller determines the green time for the next phase. During the training time, CS evaluates the performance of the controller with different parameters until finding the optimal parameters for the ANFIS controller. The cost function for the CS optimization method is considered as the average delay time of a complete run of a simulation. It means CS aimed to reduce average delay time of the whole network in finding optimal parameters Eq. 8.

$$costfunction = \frac{\sum_{i=1}^{n} d_i}{k} \tag{8}$$

where $i = 1, \ldots, n$ is the number of phases executed during the simulation time, d is the calculated delay time for each phase, and k is the number of cars released in each simulation scenario.

Two membership functions are considered for inputs named small, and large Fig. 2. These membership functions are set in the range between zero and 40. The training process of ANFIS controller is shown in Fig. 3.

IV. Experiments Environment and Results

For doing evaluation, we developed a fixed-fuzzy and fixed-time controllers as well.

To design fixed-FLC, it is needed to define a rule base. The rule base is defined in a way to consider neighbors situations

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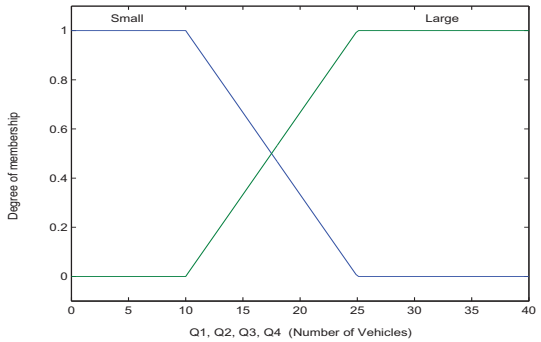


Fig. 2. The membership function definition for input of CS-ANFIS controller.

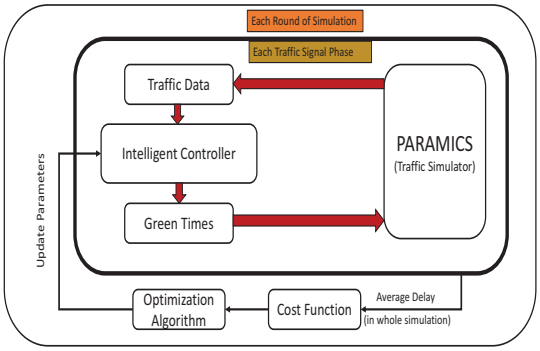


Fig. 3. The figure shows the process of ANFIS training. ANFIS parameters are updated after each round of simulation through CS optimization method.

in proposing green time. Queue length of vehicles at current link (CL), next link (NL), second next link (2NL), and third next link (3NL) are the factors considered in the rule base definition. For example we define the second rule as this:

If CL is ***medium****, NL is* ***small****, 2NL is* ***not medium****, and 3NL is* ***not medium****, then Qreen Time is* ***medium****.*

Table I shows the rule base used for both fixed-FLC. Here, we have listed 12 rows, however, most of them represent more than one rule.

In the case of fixed-FLC, the membership functions of the inputs cover range 0-40 as the input parameters. The value of inputs show the queue length. The possible green time or output of the FLC is between 0-80. A FLC with 4 inputs and one output is designed for an intersection with four approaching links.

Fig. 4 shows the initial setting of the membership functions for fixed-FLC inputs and output. Each input and the output have three membership functions named small, medium, and large. Small and large are trapezoidal functions and medium is a triangular one. The fixed-FLC works based on these settings. As it is mentioned before, we considered just two of the input membership functions for ANFIS controller Fig.

TABLE I
FLC Rule Base. In this table S stands for small, M for medium, L for large, and ∼ is for negation (eg., ∼S means not-small, which could be either medium or large).

	CL	NL	2NL	3NL	Green Time
(1)	S	S/M/L	S/M/L	S/M/L	S
(2)	M	S	∼M	∼M	S
(3)	M	M	M	M	M
(4)	M	L	S/M/L	S/M/L	M
(5)	L	∼L	∼L	∼L	M
(6)	L	L	∼L	∼L	L
(7)	L	∼L	L	∼L	M
(8)	L	∼L	∼L	L	M
(9)	L	∼L	L	L	M
(10)	L	L	∼L	L	L
(11)	L	L	L	∼L	L
(12)	L	L	L	L	L

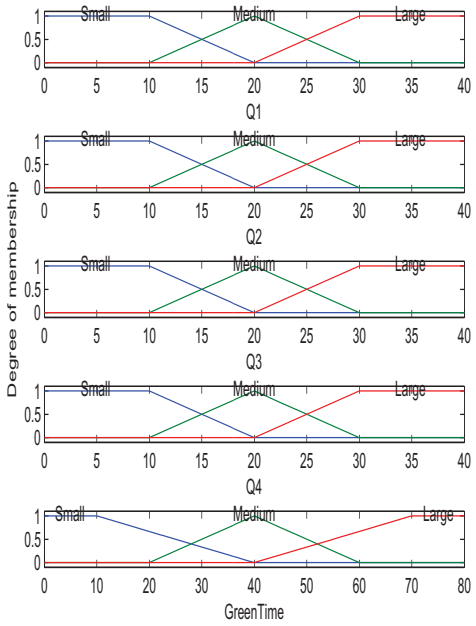


Fig. 4. Membership functions of inputs and the output of the FLC.

2. It helps to reduce the optimization time, but it also may affect the accuracy of the controller.

Fixed-time controller or pre-timed controller is a common benchmark for evaluating the performance of controllers. In designed fixed-time controller, a constant amount of time is set for each phase. This factor, reduce the flexibility of the fixed-time controller to adapt traffic demands. Deigned fixed-time controllers use equal time for all green phases. During this work, three fixed-time controllers with 20, 40, and 60 seconds values. This is done to have a more comprehensive comparison by considering different values for fixed-time controllers.

An intersection with four approaching links and four

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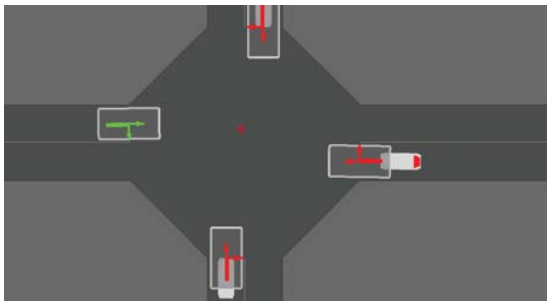


Fig. 5. Snapshot of four defined phases at an isolated intersection in Paramics.

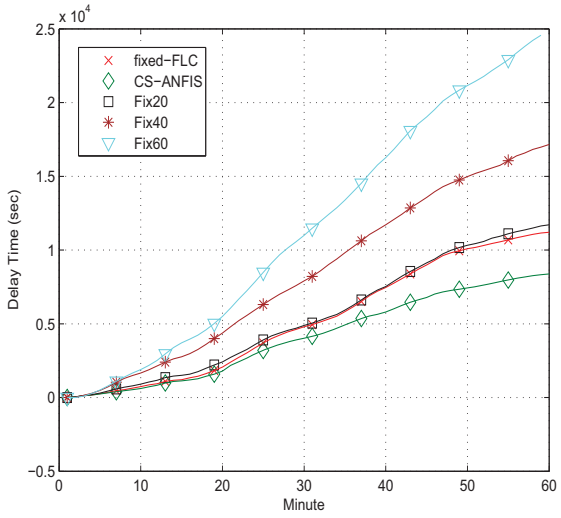


Fig. 6. Accumulative total delay time of the intersection in one hour.

phases is designed as a testing network model in Paramics V6.8.0. Fig. 5. Zones are the areas that vehicles are released from them to the intersection and we consider four zones in our simulation. During the test period, 1200 cars release to the intersection from these zones. Matlab R2011b is used to implement the controller.

The performance of developed controllers are measured by calculating the total delay time of the intersection. Fig. 7 presents the accumulative delay time of the intersection in a one-hour simulation after utilizing each controller.

The diagram illustrates that the ANFIS controller, which is optimized by CS algorithm, has the best performance by causing less amount of delay to the traffic network. It should be considered that this higher performance is obtained while we just put two membership function for each inputs in ANFIS controller. Reducing the number of membership function usually reduce its accuracy especially when the number of membership function is few. Here, this elimination is done for reducing the computational cost of the controller, but we still can see its better performance. Second best place

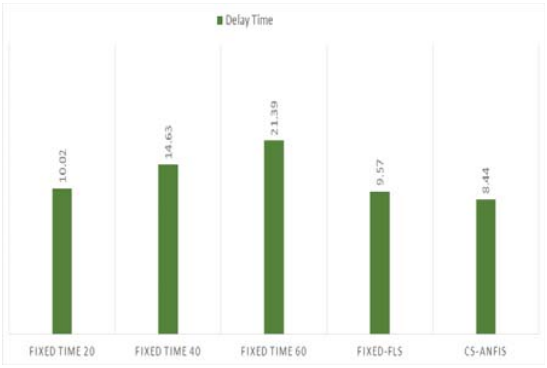


Fig. 7. Total delay time per vehicle for the intersection.

is for fixed-FLC and after all we have the fixed-time. Fixed-time controller is performing different for each green phase value. These differences prove that the efficiency of the fixed-time controller is variable and highly depends on the traffic conditions.

Fig. 7 illustrates the total delay that impose to the intersection per vehicle in one hour. Bar charts show that the best performance is for ANFIS traffic controller. It describes the amount of delay per vehicle per hour for the junction. Delays are presented in seconds. In overall, both fuzzy controllers have better performance than fixed-time controller, while the best performance is for optimized ANFIS controller used CS optimization method.

V. Conclusions

In this paper, we implement and examine the performance of ANFIS for controlling traffic signals for an isolated intersection. ANFIS gives the opportunity of using fuzzy logic system in traffic signal controlling without the need of a pre-defined rule base. The proposed controller obtains its optimal rules by using the cuckoo search algorithm (CS). The CS algorithm is designed to minimize the total delay of the traffic network. A fixed-FLC and a fixed-time controller with three different values are also designed and implemented as benchmarks for evaluating the performance of designed CS-based ANFIS controller. Trapezoidal and triangular membership functions are considered for queue lengths in fuzzy controllers. Experimental results for the simulation scenarios show an improved performance of the ANFIS controller compared to fixed-FLC and fixed-time methods.

As future work, the design and implementation of similar controllers for a multi-intersection network is planned. Considering the situation of neighbor intersections is important and should be taken into consideration in designing controllers for a multi-intersection network.

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